

Markscheme

Mathematics: analysis and approaches

Practice question bank

43. (a)

Consider the simultaneous equations $\log_2 x + \log_2 y = 7$ and $\log_2 x - \log_2 y = 1$.

Find the value of x and the value of y .

adding the equations: $2 \log_2 x = 8 \Rightarrow \log_2 x = 4$ (M1) A1

$$x = 16 \quad \text{A1}$$

subtracting the equations: $2 \log_2 y = 6 \Rightarrow \log_2 y = 3$ (M1)

$$y = 8 \quad \text{A1}$$

[5 marks]

(b)

Hence find the value of $\log_4(xy)$.

$$xy = 128 = 4^{3.5} \quad \text{(M1)}$$

$$\log_4(128) = 3.5 \quad \text{A1}$$

[2 marks]